

Probabilistic Machine Learning

Lecture 03: Linear Models — Classical and Probabilistic

Prof. Dr. Michael Granitzer

The spine of the course: every classical method is stated first in its usual form and then re-derived as maximum likelihood under a stated noise model. Reused units carry the classical account the three probabilistic units are new.

Module 1

Linear Regression, Classically

Guiding Question: Why squared error — and what exactly does "linear" forbid?

The Task: Predicting Acceleration

The world

O — vehicles

$Y = \mathbb{R}$ — time to accelerate to 100 km/h

$\gamma(o)$ — the true acceleration time of vehicle o

The model world

α — measure wheels, horsepower, weight, torque

X — those four numbers, for every vehicle

h_{Θ} — a linear function of them

Deliberately mundane. The value is that every abstract symbol from the previous unit now has a concrete referent students can argue with — especially α , where "which four numbers" is a real modelling decision.

The Data

wheels	horsepower	weight	torque	acceleration time
4	150	1350	250	9.8
4	320	1600	430	4.9
6	95	2100	210	16.2
...	...	...	...	...

The last column is y the rest is $\mathbf{x}$.

Ask which column is likely useless (wheels, in a passenger-car dataset) and what would happen if it were included anyway — an invitation to regularisation later.



The Remaining Two Pieces

Loss function

Mean squared error — the average of the squared differences between predicted and observed acceleration times.

Learning algorithm

LMS — least mean squares, which adjusts the parameters to reduce that error.

Four choices — representation, model family, loss, optimiser — and the system is fully specified. Every method in this chapter is a different set of four.

A learning problem becomes concrete the moment you can name its **task, experience and performance measure**, and its system becomes buildable the moment you can name its **representation, model, loss and optimiser**.

Close by returning to the four-box map. This unit is the template students should apply when reading any paper: identify the four boxes before judging the result.



Fitting h by Repeated Correction

LMS (least mean squares)

Initialise the weights w .

repeat pick a training example $(\mathbf{x}, y)$ compute the prediction $h_w(\mathbf{x})$ compute the error $e = y - h_w(\mathbf{x})$ update every weight: $w_i \leftarrow w_i + \eta e x_i$ **until** convergence on D

Read the update aloud in words: move each weight in the direction that would have reduced this example's error, in proportion to how much that feature contributed. That sentence is all of gradient descent, before any calculus.

What the Update Rule Encodes

Direction

The sign of e decides whether the prediction goes up or down.

Attribution

The factor x_i gives each feature credit proportional to its value.

Step size

The learning rate η trades speed against stability.

Too large an η oscillates or diverges too small an η converges slowly. It is a hyperparameter — chosen, not learned.

Speaker notes

Ask what happens with η far too large before showing it the prediction that "it overshoots and gets worse" is usually made correctly, which earns the hyperparameter concept. The visualisation in the optimisation unit shows it.



Classification versus Regression

Classification

Y is a finite set of labels.

Will this customer default — yes or no?

Regression

$Y = \mathbb{R}$, the target is a quantity.

What credit line should this customer receive?

The learning machinery is the same only the shape of Y — and therefore the loss — changes.

Keep the same running example across both to make the point that the task, not the data, decides which you are doing. The credit example returns in logistic regression, where a regression model is pressed into classification.

The Model

Linear regression predicts the target as a weighted sum of the features:

$$h_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1x_1 + \dots + w_px_p = \mathbf{w}^\top \mathbf{x}$$

with $x_0 = 1$ absorbing the intercept.

The hypothesis space is $\mathbb{R}^{p+1}$: one hypothesis per weight vector.

Point back at the hypothesis-space unit — here it is a continuous space of finite dimension, which is why calculus rather than enumeration finds the optimum.

The Objective: Residual Sum of Squares

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w} \in \mathbb{R}^{p+1}} \text{RSS}(\mathbf{w}), \quad \text{RSS}(\mathbf{w}) = \sum_{i=1}^n (y_i - \mathbf{w}^\top \mathbf{x}_i)^2$$

RSS measures **effectiveness** — how good the predictions are. Do not confuse it with **efficiency**, which is about runtime and computational effort.

The effectiveness/efficiency distinction is from the source and worth keeping; students routinely say "efficient model" when they mean "accurate model".



One Feature: The Closed Form

For a single feature the minimiser can be written down directly:

Slope

$$\hat{w}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}$$

The covariance of x and y over the variance of x .

Intercept

$$\hat{w}_0 = \bar{y} - \hat{w}_1 \bar{x}$$

The fitted line passes through the point of means.

Both formulas are built from quantities the students defined in the statistics module — covariance and variance. Say that explicitly: regression is not new mathematics, it is the old summary statistics arranged to answer a prediction question.

How Much Variance Is Explained?

$$\text{TSS} = \text{ESS} + \text{RSS}, \quad R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}$$

where TSS is the total, ESS the explained and RSS the residual sum of squares.

R^2 is the fraction of the target's variation the model accounts for: 0 means no better than predicting the mean, 1 means an exact fit.

Training R^2 never decreases when a feature is added, however useless the feature. A high training R^2 is not evidence that a model generalises — test R^2 and adjusted R^2 can and do fall.

Speaker notes

Connect to the baseline unit: $R^2 = 0$ is exactly the trivial-regression baseline. The warning is the regression counterpart of training error — and the motivation for the cross-validation the students already know.



Higher Dimensions

With p features the same objective in matrix form,

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

gives the optimum in one step whenever $\mathbf{X}^\top \mathbf{X}$ is invertible.

Invertibility fails when features are linearly dependent or when $p > n$ — more unknowns than equations. Both are common in practice, and both are why regularisation exists.

Do not derive the normal equations state them and spend the time on when they break. The $p > n$ case is the modern one — genomics, text — and motivates the next chapters more than the algebra does.



Takeaway

Linear regression fits a weighted sum by minimising squared residuals. Its solution is built from **covariance over variance**, and R^2 reports how much variation it explains — not whether it will generalise.

Linear in the Parameters, Not in the Features

A model is **linear** when it is linear in the **weights** $\mathbf{w}$. The features may be arbitrary transformations of the input.

Plain

$$h(\mathbf{x}) = w_0 + w_1x$$

a straight line

Polynomial

$$h(\mathbf{x}) = w_0 + w_1x + w_2x^2$$

a curve — still a *linear model*

Basis functions

$$h(\mathbf{x}) = \sum_j w_j \phi_j(\mathbf{x})$$

any fixed feature map ϕ

Speaker notes

This is the slide that unlocks the rest: the same least-squares machinery fits curves, interactions and periodic patterns, as long as the transformation is chosen in advance. Ask what is *not* expressible — anything where the transformation itself must be learned, which is the neural-network module.



Consequences

The good

The closed-form solution and all statistical guarantees carry over unchanged.

The catch

The feature map is a **modelling decision**. Choose too rich a basis and the model overfits too poor and it underfits.

Increasing polynomial degree is a dial on the bias–variance axis you can watch turning.

Tie back to principles: the basis is the inductive bias made explicit. This is the cleanest example in the course of choosing a hypothesis space deliberately.

The Question Nobody Asks

We minimised the *squared* residuals. Why squares — why not absolute deviations, or fourth powers?

Guiding question. Most students have never been given a reason beyond differentiability. There is a real one, and it is worth the slide.

Geometric View

The predictions $\mathbf{X}\mathbf{w}$ live in the **column space** of $\mathbf{X}$ — everything the features can express. The observed $\mathbf{y}$ generally does not.

Least squares picks the point of that space **closest to $\mathbf{y}$ in Euclidean distance**: the orthogonal projection. The residual is perpendicular to every feature.

Consequence

The residual carries no remaining linear information about the features.

Speaker notes

The orthogonality condition is the geometric content of the normal equations. It also explains why adding a feature can never increase RSS: the space you project onto only grew.



The Gauss–Markov Theorem

Gauss–Markov (informal). Among all estimators that are

- **linear** in the observed outputs $(y_1, \dots, y_n)$, and
- **unbiased**,

the least-squares estimator has the **lowest variance**. It is the *Best Linear Unbiased Estimator* — BLUE.

Best

lowest variance among the candidates

Linear

a linear function of the observations

Unbiased

right on average over repeated samples

Speaker notes

Spell out what the theorem does *not* say: nothing about non-linear or biased estimators — and biased estimators (ridge, lasso) can beat OLS in prediction error. That is not a contradiction, it is a different criterion.



What This Buys, and What It Costs

The payoff

Minimising RSS yields optimal statistical properties under **very mild assumptions** — no normality needed, only uncorrelated errors of constant variance.

The price

Alternatives — absolute error, Huber loss — are **more robust to outliers** but give up the BLUE guarantee.

Squared error is not an arbitrary convenience: it is the loss whose minimiser is the best linear unbiased estimator. Departing from it is legitimate — but it is a trade, and you should be able to name what you traded.

Close the loop with the robustness discussion from descriptive statistics: mean versus median reappears here as squared versus absolute loss. Same trade-off, new context — exactly the spiral the pedagogic concept asks for.

Two Ways to the Same Minimum

Direct (closed form)

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

Exact, one step. Requires forming and inverting a $(p+1) \times (p+1)$ matrix — costly for large p , impossible when it is singular.

Iterative (gradient descent)

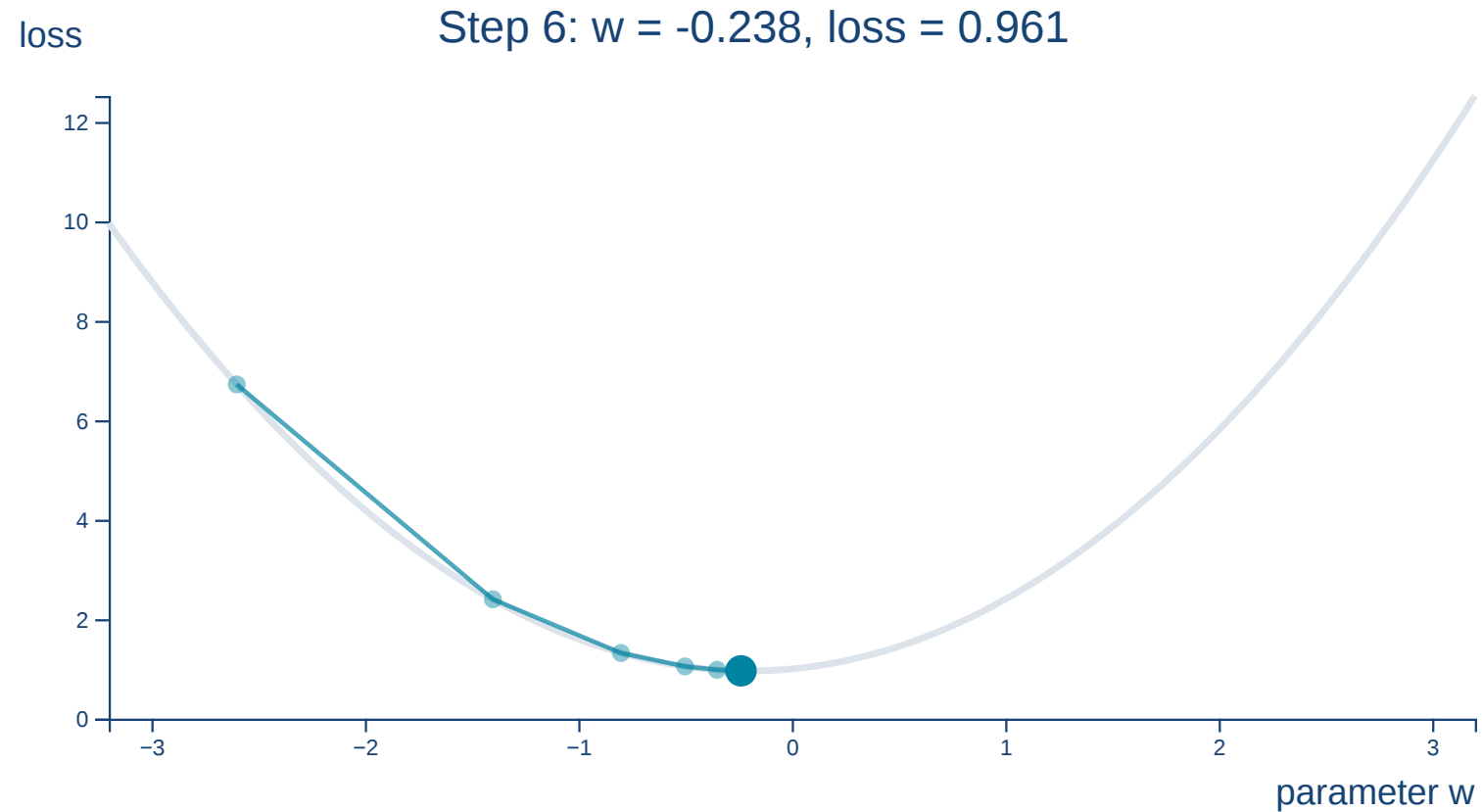
$$\mathbf{w} \leftarrow \mathbf{w} - \eta \nabla \text{RSS}(\mathbf{w})$$

Approximate, many steps. Scales to large p and n , works when the direct solution does not exist, and extends to models with no closed form at all.

The iterative rule is exactly the **LMS algorithm** from Module 2: the update $w_i \leftarrow w_i + \eta e x_i$ is a step along $-\nabla \text{RSS}$, written per example instead of per dataset.

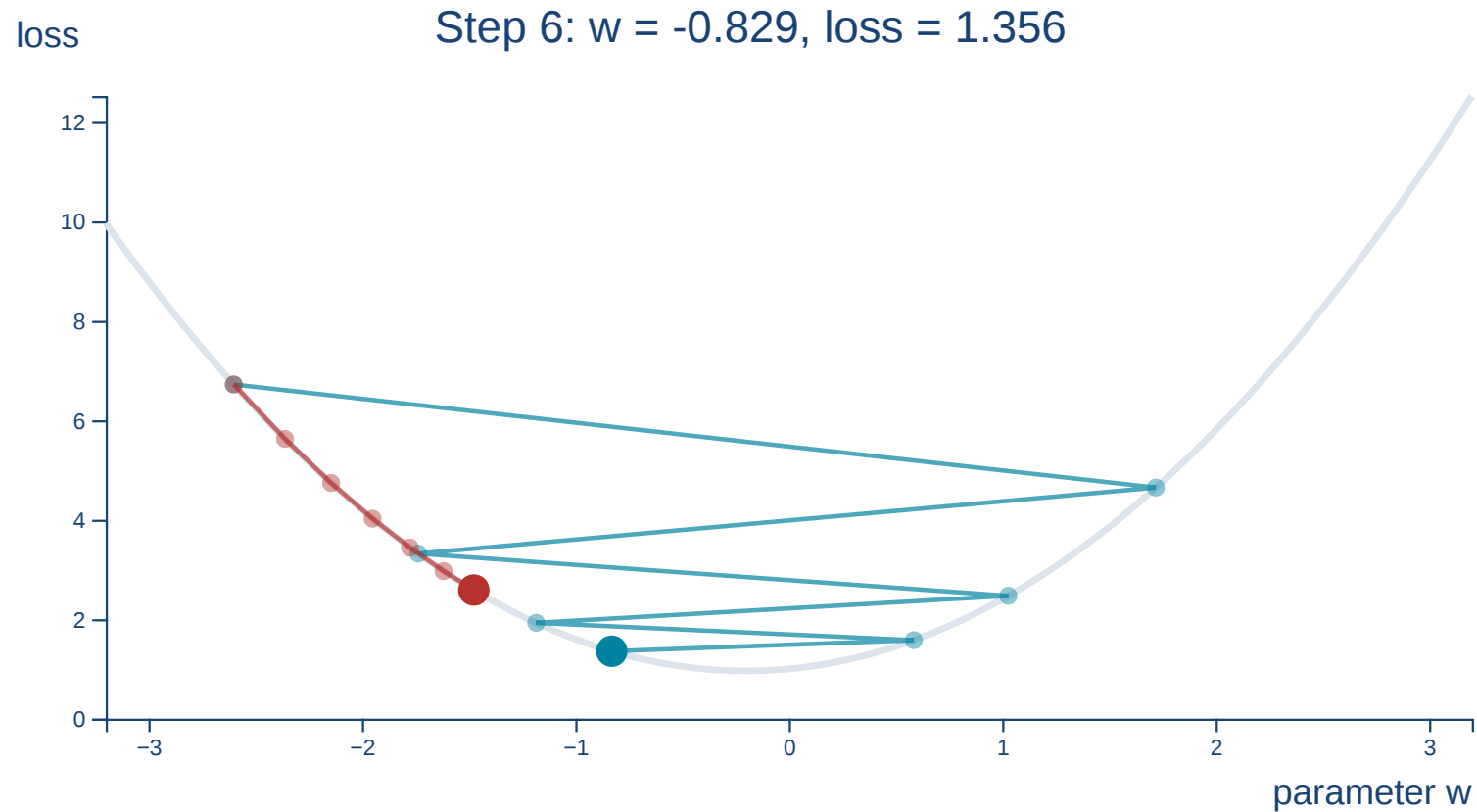
The right-hand column is the one that generalises: logistic regression and neural networks have no closed form, so the iterative method is the only one that survives the rest of the course. Make the LMS identification explicit — students otherwise carry two unconnected procedures.

Watching Gradient Descent Work



Stepped viz (P6). Each click is one update. Ask before the first click where the steps will be largest — where the slope is steepest, far from the minimum — and why they shrink as the minimum approaches, without η changing.

The Learning Rate Decides Everything



Teal: $\eta = 0.9$ — overshoots and oscillates across the minimum. Red: $\eta = 0.05$ — stable, but barely moves.



The learning rate is a **hyperparameter**: it is chosen, not learned, and both failure modes are silent — one looks unstable, the other looks converged.

Speaker notes

Predict-then-check: ask which curve belongs to which η before revealing. The "looks converged" failure is the dangerous one — a slow run and a finished run produce the same-looking loss curve over a short window.



Takeaway

Use the **closed form** when the matrix is small and well-conditioned use **gradient descent** when it is not — which, for everything after this unit, is always.

Module 2

The Probabilistic Reading

Guiding Question: *Least squares was justified after the fact. What if we had stated how the data arose first — would we get the same estimator?*

Turning a Model Into an Estimator

We have a model with parameters θ and data D . The model assigns each possible dataset a probability.

The **likelihood** is that probability, read as a function of the **parameters** with the data held fixed:

$$L(\theta) = p(D \mid \theta)$$

The **maximum likelihood estimate** is $\hat{\theta}_{\text{ML}} = \arg\max_{\theta} p(D \mid \theta)$.

$p(D \mid \theta)$ is a probability over **data**, not over parameters. The likelihood is not a distribution over θ and does not integrate to 1.

Speaker notes

The warning is the single most common confusion in the course. The likelihood is the same expression as the sampling distribution, read along the other axis. Bayesian methods later put an actual distribution on θ this is where the two approaches part.



Why the Logarithm

Under the **i.i.d. assumption** — observations independent and identically distributed — the likelihood factorises:

$$L(\theta) = \prod_{i=1}^n p(x_i \mid \theta) \quad \implies \quad \ell(\theta) = \sum_{i=1}^n \log p(x_i \mid \theta)$$

Same maximum

log is monotone, so it does not move the arg max.

Numerically safe

A product of thousands of small probabilities underflows; a sum of logs does not.

Differentiable sums

Sums differentiate term by term — gradient methods apply directly.

The i.i.d. assumption is doing real work here. Time series, spatial data and grouped data violate it, and the factorisation is then simply wrong.

Example: A Biased Coin

n flips, h heads, parameter $\theta = P(\text{heads})$:

$$\ell(\theta) = h \log \theta + (n - h) \log(1 - \theta) \quad \frac{d\ell}{d\theta} = \frac{h}{\theta} - \frac{n - h}{1 - \theta} = 0 \quad \hat{\theta}_{\text{ML}} = \frac{h}{n}$$

The estimator is the observed frequency — the answer intuition would have given. MLE **derives** it rather than assuming it, and the same derivation works where intuition has nothing to offer.

Mini-Exercise

You flip a coin **3 times** and see 3 heads.

1. What is $\hat{\theta}_{\text{ML}}$?
2. Is that a reasonable belief about the coin?
3. What is the estimator missing?

Mini-Exercise — Solution

Maximum likelihood picks the parameters under which the observed data is most probable. It is the bridge from a **model of the world** to a **learning algorithm** — and it is blind to prior knowledge and to its own uncertainty.

The three-heads example is the hook for the whole Bayesian thread: students should feel MLE's failure before regularisation and priors are offered as the fix.

Mini-Exercise — Solution

1. $\hat{\theta} = 3/3 = 1$ — the coin never comes up tails.

2. No. Three flips is no basis for excluding tails entirely.

3. Any **prior** knowledge that coins are usually near-fair, and any account of its own **uncertainty**. MLE returns a single point and no error bars.

Both gaps are what the Bayesian treatment later repairs.

Maximum likelihood picks the parameters under which the observed data is most probable. It is the bridge from a **model of the world** to a **learning algorithm** — and it is blind to prior knowledge and to its own uncertainty.

The Same Model, Told Differently

Classical account

Fit a line by **minimising squared error**:

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w}} \sum_i (y_i - \mathbf{w}^\top \mathbf{x}_i)^2$$

Justified after the fact by Gauss–Markov: it is the best linear unbiased estimator.

Probabilistic account

State how the data **arose**:

$$y = \mathbf{w}^\top \mathbf{x} + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2)$$

Then estimate $\mathbf{w}$ by maximum likelihood.

Do these give the same answer?

Question-first (P1) and the organising contrast of the lecture. Let students guess before the derivation. The answer — identical — is the point: least squares was a probabilistic model all along, with its assumptions unstated.

The Derivation

The noise model says $p(y_i \mid \mathbf{x}_i, \mathbf{w}) = \mathcal{N}(y_i \mid \mathbf{w}^\top \mathbf{x}_i, \sigma^2)$, so

$$\begin{aligned} \ell(\mathbf{w}) &= \sum_{i=1}^n \log \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - \mathbf{w}^\top \mathbf{x}_i)^2}{2\sigma^2}\right) = \text{const} - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mathbf{w}^\top \mathbf{x}_i)^2 \end{aligned}$$

Maximising the log-likelihood **is** minimising the residual sum of squares. The constant and the $1/2\sigma^2$ do not move the arg max. $\hat{\mathbf{w}}_{\text{ML}} = \hat{\mathbf{w}}_{\text{OLS}}$

Speaker notes

Walk the two lines slowly — this is the moment the course's thesis becomes concrete. The squared term falls out of the *exponent of the Gaussian*: squared error is not a choice, it is what a Gaussian noise assumption implies.



What the Probabilistic View Adds

The assumptions become visible

Additive noise, **zero mean**, **constant variance** (homoscedasticity), **independent** across observations, **Gaussian**.

They become checkable

Plot the residuals. Fanning out? Variance is not constant. Skewed? Not Gaussian. Patterned? Not independent.

You get a distribution

Not just $\hat{y}$ but $p(y \mid \mathbf{x})$ — a **predictive distribution** with an interval, not a point.

Choosing a loss and choosing a noise model are the **same decision**. Squared error $\Leftrightarrow$ Gaussian noise absolute error $\Leftrightarrow$ Laplace noise, which is why it is more robust to outliers.

Speaker notes

The loss/noise correspondence is the transferable idea, and it explains the robustness result from the Gauss–Markov unit rather than merely asserting it: the Laplace distribution has heavier tails, so large residuals are less surprising and pull the fit less.



Mini-Exercise

You fit a linear model to house prices. The residual plot shows the spread of errors **growing** with the predicted price.

1. Which assumption is violated?
2. Is $\hat{\mathbf{w}}$ still meaningful?
3. What is definitely wrong?

Mini-Exercise — Solution

Least squares **is** maximum likelihood under Gaussian noise. Stating the model probabilistically changes no estimate — it exposes the assumptions and buys you a predictive distribution.

Mini-Exercise — Solution

1. Homoscedasticity — the noise variance is not constant.

2. The point estimate remains unbiased it is just no longer the minimum-variance one.

3. Every **interval** and every **p-value**: they are computed from a single σ that does not exist. Predictions are over-confident for expensive houses and under-confident for cheap ones.

Least squares **is** maximum likelihood under Gaussian noise. Stating the model probabilistically changes no estimate — it exposes the assumptions and buys you a predictive distribution.

Module 3

From Regression to Classification

Guiding Question: A decision is a yes or a no. What goes wrong when we fit a straight line to one?

Fitting a Line to Labels

Encode the two classes as $y \in \{-1, +1\}$, fit a linear model, and classify by the sign of the prediction:

$$h(\mathbf{x}) = \text{sign}(\mathbf{w}^\top \mathbf{x})$$

One feature

The **discrimination point** x' satisfies

$$w_0 + w_1 x' = 0$$

Everything to one side is one class.

Two features

The **discrimination line** satisfies

$$w_0 + w_1 x_1 + w_2 x_2 = 0$$

In p dimensions it becomes a hyperplane.

The decision boundary is where the model output crosses zero — a level set. Say it once here it holds for logistic regression, SVMs and a neural network's final layer alike.

Where It Breaks

Squared loss punishes correctness

A point far on the *correct* side has a large residual, so least squares drags the boundary toward it.

The boundary shifts

Adding well-classified points — or an outlier — moves the discrimination point, even when nothing was misclassified.

The output is also unbounded: $\mathbf{w}^\top \mathbf{x}$ can be 17 or -40 , and neither is interpretable as a degree of belief.

Speaker notes

Force the interpretation (P2): ask why "being very right" should ever count as error. Once students see that squared loss is the wrong loss for a decision, the sigmoid arrives as a fix rather than a formula.



Two Problems, One Fix

$\text{sign}(\mathbf{w}^\top \mathbf{x})$ is not differentiable

No gradient means no gradient descent — the optimisation machinery does not apply.

The boundary is unstable

Squared loss on ± 1 labels lets well-classified points shift the discrimination point.

Replace the hard threshold with the **logistic sigmoid**:

$$h_{\Theta}(\mathbf{x}) = \text{expit}(\mathbf{w}^\top \mathbf{x}) \approx P(y = 1 \mid \mathbf{x})$$

The output becomes a smooth, bounded quantity that can be read as a probability — and can be differentiated.

Both problems have the same root: a decision is a step function, and steps have no useful derivative. Softening the step is the standard move; it returns as the activation function in neural networks.

Odds and Log-Odds

Odds

The ratio of favourable to remaining cases.

Rolling a six: odds 1 : 5.

Logit

The **log-odds** of a probability p :

$$\text{logit}(p) = \ln \frac{p}{1 - p}$$

Range

$p \in (0, 1)$ maps to $(-\infty, +\infty)$
— exactly the range a linear model produces.

The modelling assumption: the logit of the target is a linear combination of the features.

$$\ln \frac{P(y = 1 \mid \mathbf{x})}{1 - P(y = 1 \mid \mathbf{x})} = \mathbf{w}^\top \mathbf{x}$$

Speaker notes

This is the cleanest way in: we do not force a probability to be linear (it cannot be — it is bounded), we make its log-odds linear. The sigmoid is then not an arbitrary squashing function but the inverse of the logit.



The Model

Inverting the logit gives the **logistic sigmoid**:

$$h_{\Theta}(\mathbf{x}) = \text{expit}(\mathbf{w}^{\top} \mathbf{x}) = \frac{1}{1 + e^{-\mathbf{w}^{\top} \mathbf{x}}}$$

Shape

An S-curve: near-linear around 0, saturating towards 0 and 1 at the extremes.

Reading

$h_{\Theta}(\mathbf{x}) = P(y = 1 \mid \mathbf{x})$, and the decision boundary is still $\mathbf{w}^{\top} \mathbf{x} = 0$ — where the sigmoid passes 0.5.

Speaker notes

The boundary is unchanged from linear regression for classification what changed is the *loss* and the interpretability of the output. Worth stating explicitly, since students often expect a curved boundary.



The Loss: Cross-Entropy

Binary cross-entropy (log loss)
$$L(\Theta) = -\frac{1}{n} \sum_{i=1}^n \left[y_i \ln h_{\Theta}(\mathbf{x}_i) + (1 - y_i) \ln (1 - h_{\Theta}(\mathbf{x}_i)) \right]$$

For $|Y| > 2$ classes, the multiclass form sums over the classes with a softmax output.

Where does it come from? From the **likelihood**: $P(y_i = 1 \mid \mathbf{x}_i; \Theta) = h_{\Theta}(\mathbf{x}_i)$ and $P(y_i = 0 \mid \mathbf{x}_i; \Theta) = 1 - h_{\Theta}(\mathbf{x}_i)$. Maximising the likelihood of the data is minimising this loss.

Speaker notes

The derivation is one line and worth doing aloud: write the likelihood of one example as $h^y (1-h)^{(1-y)}$, take the product over examples, take the log, negate. Maximum likelihood is the origin of the loss — not a separate idea.



Why It Fixes the Problem

Confident and correct

Loss goes to zero — no penalty for being emphatically right.

Confident and wrong

Loss goes to infinity — exactly the errors we want punished.

There is no closed-form solution the parameters are found by gradient descent on this loss.

Contrast directly with the squared-loss failure two units back: cross-entropy punishes confident mistakes and ignores confident successes, which is what a classification loss should do.

Takeaway

Logistic regression is a **linear model of the log-odds**, trained by maximising likelihood. It keeps the linear decision boundary but produces a calibrated probability — and a loss that punishes the right kind of error.

Module 4

Logistic Regression, Probabilistically

*Guiding Question: Cross-entropy was chosen because it punishes confident mistakes.
Where does it actually come from?*

The Same Pattern Again

Classical account

Squash a linear score with a sigmoid, then minimise **cross-entropy** because it punishes confident mistakes.

Probabilistic account

State the generative story: the label is a **Bernoulli** draw,

$$y \mid \mathbf{x} \sim \text{Bernoulli}(\sigma(\mathbf{w}^\top \mathbf{x}))$$

then maximise the likelihood.

The Derivation

With $h_i = \sigma(\mathbf{w}^\top \mathbf{x}_i)$, one observation has probability $h_i^{y_i} (1 - h_i)^{1-y_i}$, so

$$\ell(\mathbf{w}) = \sum_{i=1}^n \left[y_i \log h_i + (1 - y_i) \log(1 - h_i) \right] = -n \cdot (\text{cross-entropy})$$

Cross-entropy **is** the negative log-likelihood of a Bernoulli model. As with squared error and the Gaussian, the loss was never an independent choice.

Gaussian noise

→ squared error → linear regression

Bernoulli labels

→ cross-entropy → logistic regression

Speaker notes

Two instances of one pattern, side by side. Ask what loss a **Poisson** model would imply for count data — the students can now derive it rather than look it up, which is the transferable skill.



What It Adds: Calibration

The output is now a **claim about frequencies**: among all cases predicted at 0.7, roughly 70% should be positive.

Calibrated

A reliability plot — predicted probability against observed frequency — lies on the diagonal. The number can be fed into a cost-based decision.

Miscalibrated

Systematically over- or under-confident. Accuracy can still be high the probabilities are simply not usable as probabilities.

Accuracy does not test calibration and calibration does not test accuracy. A model that outputs the base rate for everyone is perfectly calibrated and useless.

Regularisation Is a Prior

Adding an L_2 penalty to the loss:

$$\arg\max_{\mathbf{w}} \underbrace{\log p(D \mid \mathbf{w})}_{\text{likelihood}}$$

- $\underbrace{\log p(\mathbf{w})}_{\text{prior}} \propto p(\mathbf{w}) = \mathcal{N}(0, \tau^2 \mathbf{I})$

L_2 regularisation is exactly a **zero-mean Gaussian prior** on the weights, and the regularised estimate is the **MAP** estimate. L_1 corresponds to a Laplace prior, whose peak at zero is why it produces sparsity.

Cross-entropy is the Bernoulli negative log-likelihood **regularisation is a prior** and the probabilistic reading turns two heuristics into one principle — maximise likelihood, optionally times a prior.

Speaker notes

This slide retires two pieces of folklore at once. Students who have met "L2 shrinks weights" and "L1 gives sparsity" as facts now have the reason. It also sets up Bayesian linear regression, where the prior stays a distribution instead of collapsing to a point.



